

# Hengrong Du

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| CONTACT INFORMATION | Fisk University<br>Department of Math & Computer Science                                                                                                        | Page: <a href="https://hengrongdu.netlify.app/">https://hengrongdu.netlify.app/</a><br>E-mail: <a href="mailto:hdu@fisk.edu">hdu@fisk.edu</a> |
| RESEARCH INTERESTS  | Partial Differential Equations for Machine Learning and Complex Fluids                                                                                          |                                                                                                                                               |
| ACADEMIC POSITIONS  | <b>Fisk University</b> , Nashville, TN, USA<br>Assistant Professor, <a href="#">Department of Mathematics and Computer Science</a>                              | August 2025-Present                                                                                                                           |
|                     | <b>University of California, Irvine</b> , Irvine, CA, USA<br>Visiting Assistant Professor, <a href="#">Department of Mathematics</a>                            | July 2024-June 2025                                                                                                                           |
|                     | <b>Vanderbilt University</b> , Nashville, TN, USA<br>Postdoc Scholar (Research), <a href="#">Department of Mathematics</a><br>Mentor: <b>Dr. Gieri Simonett</b> | Sep. 2021-June 2024                                                                                                                           |
| EDUCATION           | <b>Purdue University</b> , West Lafayette, IN<br>Ph.D., Mathematics<br>Advisor: <b>Dr. Changyou Wang</b>                                                        | Aug. 2016-Aug. 2021                                                                                                                           |
|                     | <b>Beijing Normal University</b> , Beijing, China<br>M.S., Computational Mathematics<br>Advisor: <b>Dr. Jiequan Li</b>                                          | Sep. 2013-July 2016                                                                                                                           |
|                     | <b>South China Normal University</b> , Guangzhou, China<br>B.S., Mathematics and Applied Mathematics                                                            | Sep. 2009-July 2013                                                                                                                           |

## PUBLICATIONS

### Diffusion and Optimal Transport in Machine Learning

1. Xinyang Liu, Hengrong Du, Wei Deng, Ruqi Zhang. *Optimal Stochastic Trace Estimation in Generative Modeling*. Proceedings of the 28th International Conference on Artificial Intelligence and Statistics (AISTATS 2025).
2. Yikun Bai, Rocio Diaz Martin, Hengrong Du, Ashkan Shahbazi, Soheil Kolouri. *Linear Partial Gromov–Wasserstein Embedding*. Proceedings of the 13th International Conference on Learning Representations (ICLR 2025).
3. Yikun Bai, Rocio Diaz Martin, Hengrong Du, Ashkan Shahbazi, Soheil Kolouri. *Partial Gromov–Wasserstein Metric*. Proceedings of the 13th International Conference on Learning Representations (ICLR 2025).

4. Haoyang Zheng, Hengrong Du, Qi Feng, Wei Deng, Guang Lin. *Constrained Exploration via Reflected Replica Exchange Stochastic Gradient Langevin Dynamics*. Proceedings of the 42nd International Conference on Machine Learning (ICML 2024).
5. Wei Deng, Yu Chen, Nicole Tianjiao Yang, Hengrong Du, Qi Feng, Ricky T. Q. Chen. *Reflected Schrödinger Bridge for Constrained Generative Modeling*. Proceedings of the 41st Conference on Uncertainty in Artificial Intelligence (UAI 2024).
6. Wei Deng, Yu Chen, Nicole Tianjiao Yang, Hengrong Du, Qi Feng, Ricky T. Q. Chen. *On Convergence of Approximate Schrödinger Bridge with Bounded Cost*. Proceedings of the ICML Workshop on New Frontiers in Learning, Control, and Dynamical Systems (2023).

### Complex Fluids

7. Hengrong Du, Gieri Simonett, Yuanzhen Shao. *On a thermodynamically consistent model for magnetoviscoelastic fluids in 3D*. J. Evol. Equ. 24 (2024), no. 9, 1-51.
8. Hengrong Du, Yuanzhen Shao, Gieri Simonett. *Well-posedness for magneto-viscoelastic fluids in 3D*. Nonlinear Anal. Real World Appl. 69 (2023), no. 103759.
9. Hengrong Du, Tao Huang, Changyou Wang. *Weak compactness property of simplified nematic liquid crystal flows in dimension two*. Math. Z. 302 (2022), no. 2, 2111-2130.
10. Hengrong Du, Changyou Wang. *Global weak solutions to the stochastic Ericksen–Leslie system in dimension two*. Discrete Contin. Dyn. Syst. 42 (2022), no. 5, 2175-2197.
11. Hengrong Du, Changyou Wang. *Partial regularity of a nematic liquid crystal model with kinematic transport effects*. Nonlinearity 34 (2021), no. 5, 3001-3045.
12. Hengrong Du, Yimei Li, Changyou Wang. *Weak solutions of non-isothermal nematic liquid crystal flow in dimension three*. J. Elliptic Parabol. Equ. 6 (2020), no. 1, 71-98.
13. Hengrong Du, Xianpeng Hu, Changyou Wang. *Suitable Weak Solutions for the Co-rotational Beris–Edwards System in Dimension Three*. Arch. Ration. Mech. Anal. 238 (2020), no. 2, 749-803.

## Calculus of Variations

14. Hengrong Du, Nung Kwan Yip. *Stability of self-similar solutions to geometric flows*. Interfaces Free Bound. 25 (2023), no. 2, 155–191.
15. Hengrong Du, Qinfeng Li, Changyou Wang. *Compactness of  $M$ -uniform domains and optimal thermal insulation problems*. Adv. Calc. Var. 16 (2023), no. 1, 17-43.

## PREPRINTS

1. Hengrong Du, Qi Feng, Changwei Tu, Xiaoyu Wang, Lingjiong Zhu. *Non-Reversible Langevin Algorithms for Constrained Sampling*. [arXiv:2501.11743](#), 35 pp.
2. Yikun Bai, Huy Tran, Hengrong Du, Xinran Liu, Soheil Kolouri. *Fused Partial Gromov-Wasserstein for Structured Objects*. [arXiv:2401.03662](#), 54 pp.
3. Hengrong Du, Chuntian Wang. *Partial regularity for the three-dimensional stochastic Ericksen–Leslie Equations*. [arXiv:2401.03662](#), 50 pp.

## AWARDS

- Bilsland Dissertation Fellowship June 2020-May 2021  
It was intended to give the most accomplished final-year PhD candidates an opportunity to complete the dissertation within the 2020–21 academic year by devoting full-time effort to research and writing.

## TALKS

- Southern California Applied Math Symposium  
University of California, Riverside April 2025
- PDE & Applied Mathematics seminar  
University of California, Riverside January 2025
- PDEs & Applications Seminar  
Queen’s University January 2025
- Applied Math Seminar  
The University of Alabama October 2024
- PDEs and Fluid Mechanics and Atmospheric Sciences  
West Virginia University April 2024
- AMS Special Session on Fluids: Analysis, Applications, and Beyond  
Florida State University March 2024
- Financial Mathematics Seminar  
Florida State University March 2024
- AMS Special Section on Dynamics and Regularity of PDEs  
Joint Mathematics Meeting January 2024
- Seminar on Analysis and Stochastic Analysis  
Auburn University November 2023

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|-----------------------------|-----------------------------------------------------------------------------------------------------------------|------------------|
|                             | • PDE Seminar<br>University of Tennessee, Knoxville                                                             | October 2023     |
|                             | • PDE/Applied Math Seminar<br>Indiana University, Bloomington                                                   | October 2023     |
|                             | • Richard F. Barry Jr. Seminar Series<br>Old Dominion University                                                | September 2023   |
|                             | • The 13th AIMS Conference<br>University of North Carolina Wilmington                                           | June 2023        |
|                             | • AMS Spring Central Sectional Meeting<br>University of Cincinnati                                              | April 2023       |
|                             | • Undergraduate Seminar in Mathematics<br>Vanderbilt University                                                 | March 2023       |
|                             | • Analysis Seminar<br>Wayne State University                                                                    | March 2023       |
|                             | • 3rd Biennial Meeting of SIAM Pacific Northwest Section<br>Washington State University                         | May 2022         |
|                             | • AMS Spring Central Virtual Sectional Meeting<br>Purdue University                                             | March 2022       |
|                             | • Graduate Online Mini-course<br>Beijing Normal University                                                      | November 2021    |
|                             | • PDE Seminar Fall 2021<br>Vanderbilt University                                                                | September 2021   |
|                             | • Analysis Seminar<br>The University of Alabama                                                                 | February 2021    |
|                             | • PDE Seminar<br>Purdue University                                                                              | September 2020   |
|                             | • Student Analysis Seminar<br>Purdue University                                                                 | February 2020    |
|                             | • Student Analysis Seminar<br>Purdue University                                                                 | February 2020    |
|                             | • PDE Seminar<br>Purdue University                                                                              | January 2020     |
| CONFERENCE<br>PARTICIPATION | • Tennessee Quantum Materials Workshop<br>Tennessee State University                                            | Nov. 7, 2025     |
|                             | • Lean for PDEs, an ICARM & SLMATH collaboration (Virtual)<br>The Simons Laufer Mathematical Sciences Institute | Oct. 6-9, 2025   |
|                             | • Summer School on Theory and Applications of Elliptic PDE<br>University of California, Irvine                  | June 23-27, 2025 |
|                             | • Rivière-Fabes Symposium on Analysis and PDE<br>University of Minnesota                                        | Apr. 19-21, 2023 |
|                             | • Shanks Workshop on Advances in Mathematical and Theoretical Biology<br>Vanderbilt University                  | Mar. 17-19, 2023 |
|                             | • Workshop on Geometry and Analysis of Fluid Flows                                                              |                  |

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|------------------------|---------------------------------------------------------------------|-----------------------|
|                        | Stony Brook University                                              | Jan. 16-20, 2023      |
|                        | • AMS Fall Central Sectional Meeting                                |                       |
|                        | University of Texas at El Paso                                      | Sep. 17-18, 2022      |
|                        | • Shanks Workshop on Mathematical Aspects of Fluid Dynamics         |                       |
|                        | Vanderbilt University                                               | Feb. 19-20, 2022      |
|                        | • KUMUNU-ISU Conference on PDE, Dynamical Systems, and Applications |                       |
|                        | University of Nebraska-Lincoln                                      | Oct. 23-24, 2021      |
|                        | • AMS Fall Western Sectional Meeting                                |                       |
|                        | Online                                                              | Oct. 23-24, 2021      |
|                        | • The 84th Midwest PDE Seminar                                      |                       |
|                        | Illinois Institute of Technology                                    | Oct. 26-27, 2019      |
|                        | • Midwest Geometry Conference 2019                                  |                       |
|                        | Iowa State University                                               | Sept. 6-8, 2019       |
|                        | • The 83rd Midwest PDE Seminar                                      |                       |
|                        | Indiana University Bloomington                                      | Mar. 30-31, 2019      |
| TEACHING<br>EXPERIENCE | Instructor at Fisk University                                       | Fall 2025-Present     |
|                        | • MATH 125 Discrete Mathematics                                     | Fall 2025             |
|                        | • MATH 130 Calculus II                                              | Fall 2025             |
|                        | Instructor at University of California, Irvine                      | Fall 2024-Spring 2025 |
|                        | • MATH 3A Introduction to Linear Algebra                            | Spring 2025           |
|                        | • MATH 112B Introduction to Partial Differential Equation           | Winter 2025           |
|                        | • MATH 3D Elementary Differential Equations                         | Fall 2024             |
|                        | Instructor at Vanderbilt University                                 | Fall 2021-Spring 2024 |
|                        | • MATH 3100 Introduction to Analysis                                | Spring 2024           |
|                        | • MATH 2400 Differential Equations with Linear Algebra              | Fall 2023             |
|                        | • MATH 3100 Introduction to Analysis                                | Spring 2023           |
|                        | • MATH 3100 Introduction to Analysis                                | Fall 2022             |
|                        | • MATH 1301 Accelerated Single-Variable Calculus II                 | Fall 2022             |
|                        | • MATH 2420 Methods of Ordinary Differential Equations              | Spring 2022           |
|                        | • MATH 2400 Differential Equations with Linear Algebra              | Spring 2022           |
|                        | Recitation Instructor at Purdue University                          | Fall 2016-Summer 2021 |
|                        | • MA 26200 Linear Algebra And Differential Equations                | Fall 2019             |
|                        | • MA 16200 Plane Analytic Geometry And Calculus II                  | Spring 2019           |
|                        | • MA 16500 Plane Analytic Geometry And Calculus I                   | Fall 2018             |